# RocketPy R8 — V3

Last updated: **20 September 2026**.

Weather: **18 October 2026, 14:00 Europe/Lisbon (13:00 UTC)**, location **39.388984, −8.288272**. NOAA NCEP/NCAR climate scenario for 1991–2020. Method and limitations: WEATHER_UPDATE.md.

In [1]:
```
from simulation_r8 import run_simulation, print_summary

# Default: the supplied NOAA vertical climate profile for 1991-2020.
# No wind: run_simulation(wind_speed=0.0, wind_direction_from=0.0)
env, motor, rocket, flight = run_simulation()
print_summary(flight)
```

```
Rocket:                R8
Surface wind:          1.2 m/s from 233 deg
Launch rail length:    12.00 m
Loaded mass:           51.014 kg
Initial CG:            2.636 m
Apogee AGL:            2813.9 m
Time to apogee:        25.476 s
Out-of-rail velocity: 30.693 m/s
Maximum speed:         224.196 m/s
Impact velocity:       -6.555 m/s
Impact distance:       847.0 m

EuRoC checks
Rail velocity >= 30.0 m/s: True
Rail stability >= 1.5 cal: True
Impact distance <= 2500 m: True
```

In [2]:
```
# Full set of plots and information — run if required.
flight.all_info()
```

Initial Conditions

Initial time: 0.000 s
Position - x: 0.00 m | y: 0.00 m | z: 160.00 m
Velocity - Vx: 0.00 m/s | Vy: 0.00 m/s | Vz: 0.00 m/s
Attitude (quaternions) - e0: -0.998 | e1: -0.035 | e2: -0.039 | e3: 0.026
Euler Angles - Spin ϕ : 130.00° | Nutation θ: -6.00° | Precession ψ: 227.00°
Angular Velocity - ω1: 0.00 rad/s | ω2: 0.00 rad/s | ω3: 0.00 rad/s
Initial Stability Margin: 2.617 c

Surface Wind Conditions

Frontal Surface Wind Speed: 0.21 m/s
Lateral Surface Wind Speed: 1.21 m/s

Launch Rail

Launch Rail Length: 12.0 m
Launch Rail Inclination: 84.00°
Launch Rail Heading: 133.00°

Rail Departure State

Rail Departure Time: 0.722 s
Rail Departure Velocity: 30.693 m/s
Rail Departure Stability Margin: 2.711 c
Rail Departure Angle of Attack: 2.295°
Rail Departure Thrust-Weight Ratio: 5.835
Rail Departure Reynolds Number: 3.093e+05

Burn out State

Burn out time: 6.454 s
Altitude at burn out: 1067.757 m (ASL) | 907.757 m (AGL)
Rocket speed at burn out: 222.833 m/s
Freestream velocity at burn out: 223.140 m/s
Mach Number at burn out: 0.660
Kinetic energy at burn out: 1.035e+06 J

Apogee State

Apogee Time: 25.476 s
Apogee Altitude: 2973.904 m (ASL) | 2813.904 m (AGL)
Apogee Freestream Speed: 26.001 m/s
Apogee X position: 420.154 m
Apogee Y position: -539.354 m
Apogee latitude: 39.3848483°
Apogee longitude: -8.2840740°

Parachute Events

Parachute: Pilot Parachute
	Ejection time: 25.480 s
	Inflation time: 26.480 s
	Freestream speed at inflation: 27.541 m/s
	Altitude at inflation: 2969.144 m (ASL) | 2809.144 m (AGL)

Parachute: Main Parachute
        Ejection time: 131.040 s
        Inflation time: 131.040 s
        Freestream speed at inflation: 21.977 m/s
        Altitude at inflation: 559.954 m (ASL) | 399.954 m (AGL)

Impact Conditions

Time of impact: 190.752 s
X impact: 825.720 m
Y impact: -188.689 m
Altitude impact: 160.000 m (ASL) | -0.000 m (AGL)
Latitude: 39.3880023°
Longitude: -8.2793535°
Vertical velocity at impact: -6.555 m/s
Number of parachutes triggered until impact: 2

Stability Margin

Initial Stability Margin: 2.617 c at 0.00 s
Out of Rail Stability Margin: 2.711 c at 0.72 s
Maximum Stability Margin: 3.940 c at 6.46 s
Minimum Stability Margin: 2.617 c at 0.00 s

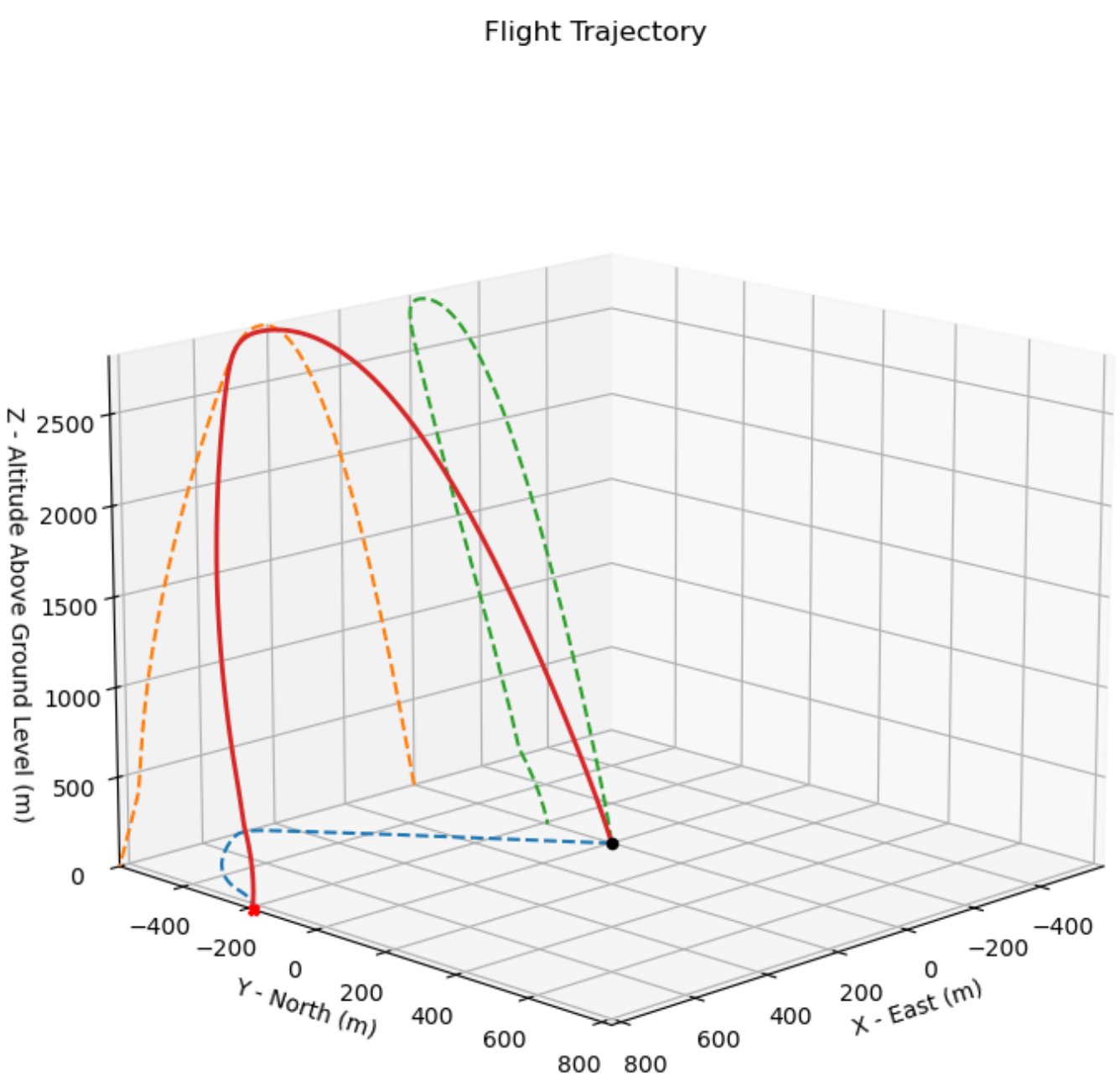
Maximum Values

Maximum Speed: 224.196 m/s at 5.94 s
Maximum Mach Number: 0.663 Mach at 5.99 s
Maximum Reynolds Number: 2.137e+06 at 5.52 s
Maximum Dynamic Pressure: 2.800e+04 Pa at 5.65 s
Maximum Acceleration During Motor Burn: 62.671 m/s² at 0.22 s
Maximum Gs During Motor Burn: 6.391 g at 0.22 s
Maximum Acceleration After Motor Burn: 68.028 m/s² at 131.04 s
Maximum Gs After Motor Burn: 6.937 Gs at 131.04 s
Maximum Stability Margin: 3.940 c at 6.46 s
Maximum Upper Rail Button Normal Force: 0.805 N
Maximum Upper Rail Button Shear Force: 0.135 N
Maximum Lower Rail Button Normal Force: 3.170 N
Maximum Lower Rail Button Shear Force: 0.544 N

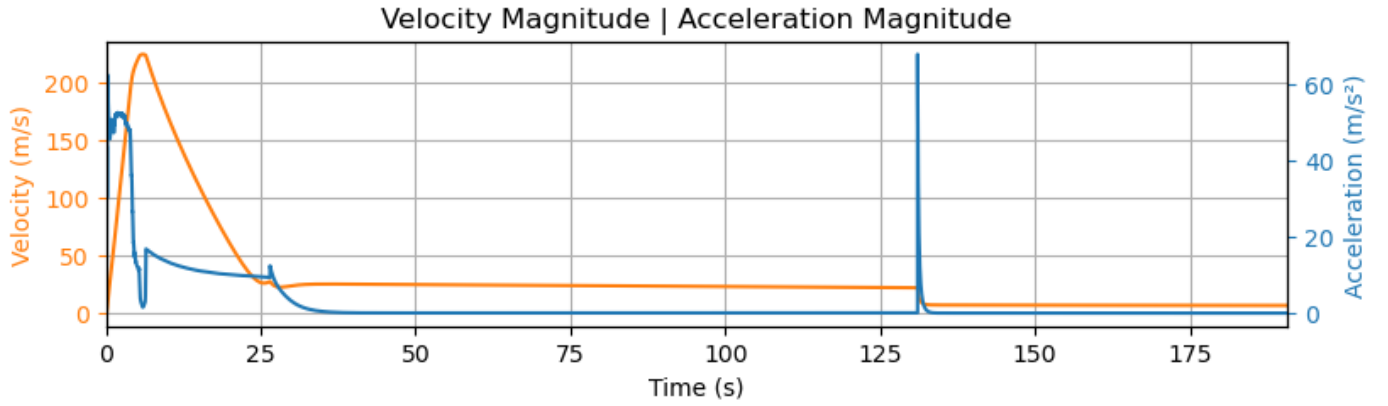
Numerical Integration Settings

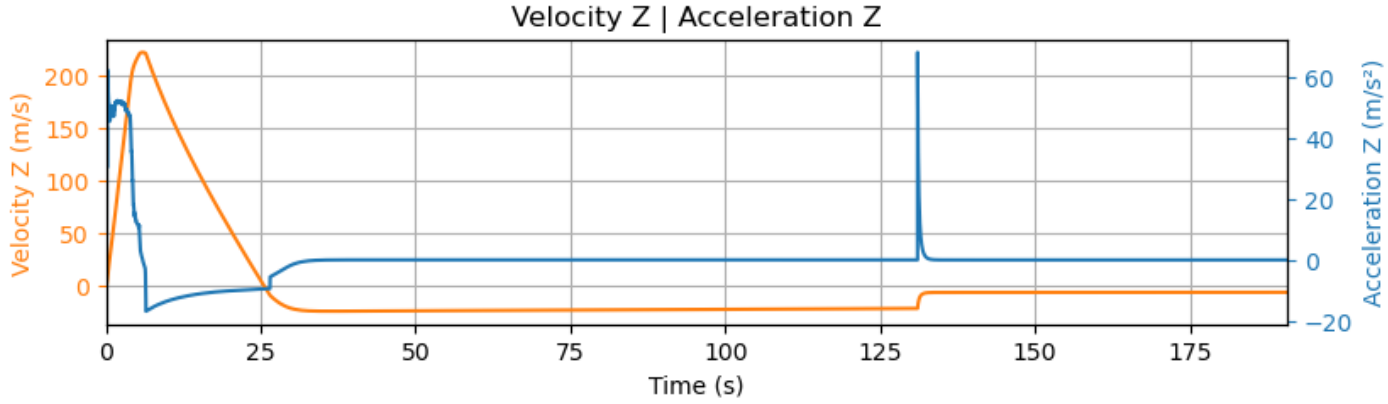
Maximum Allowed Flight Time: 600.00 s
Maximum Allowed Time Step: inf s
Minimum Allowed Time Step: 0.00e+00 s
Relative Error Tolerance: 1e-06
Absolute Error Tolerance: [0.001, 0.001, 0.001, 0.001, 0.001, 0.001, 1e-06, 1e-06, 1e-06, 1e-06, 0.001, 0.001, 0.001]
Allow Event Overshoot: True
Terminate Simulation on Apogee: False
Number of Time Steps Used: 766
Number of Derivative Functions Evaluation: 220
Average Function Evaluations per Time Step: 0.287

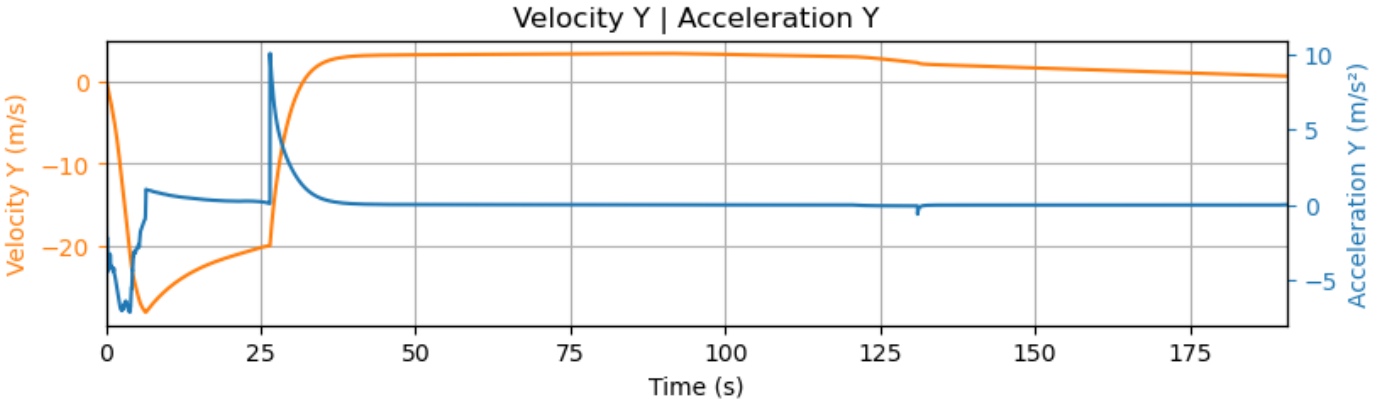
Trajectory 3d Plot

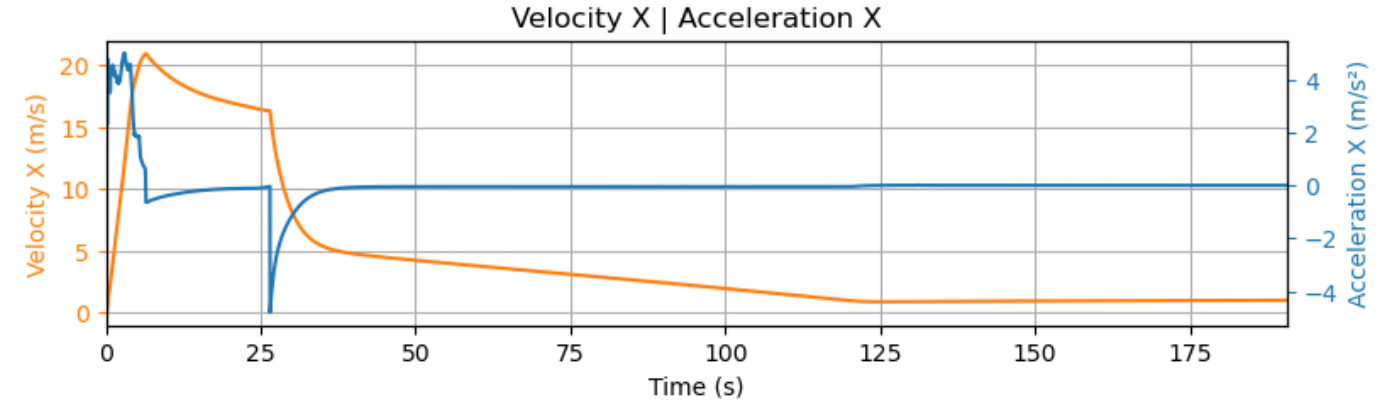



Trajectory Kinematic Plots

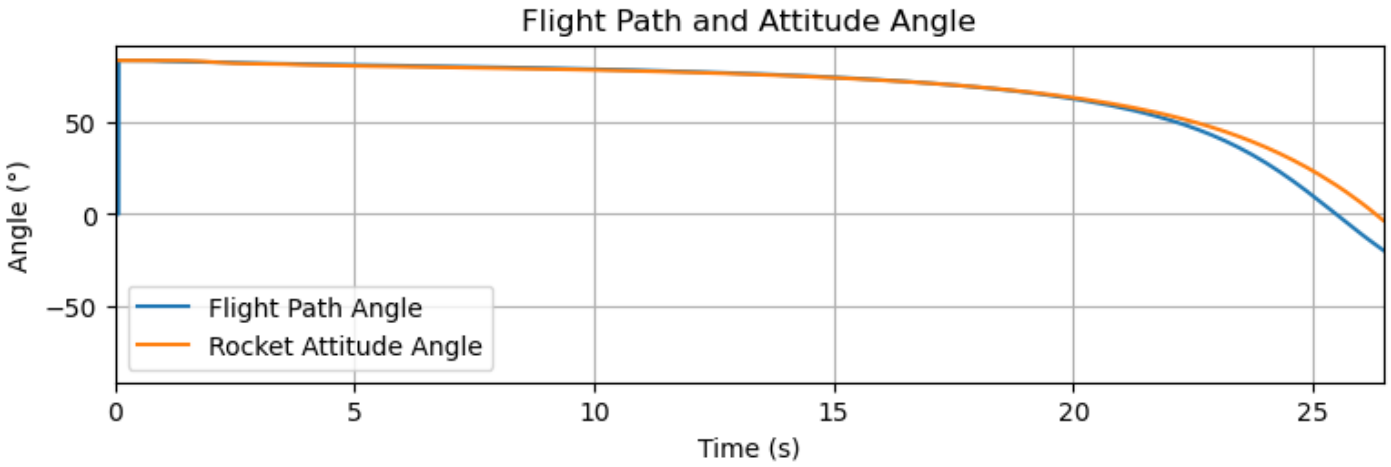


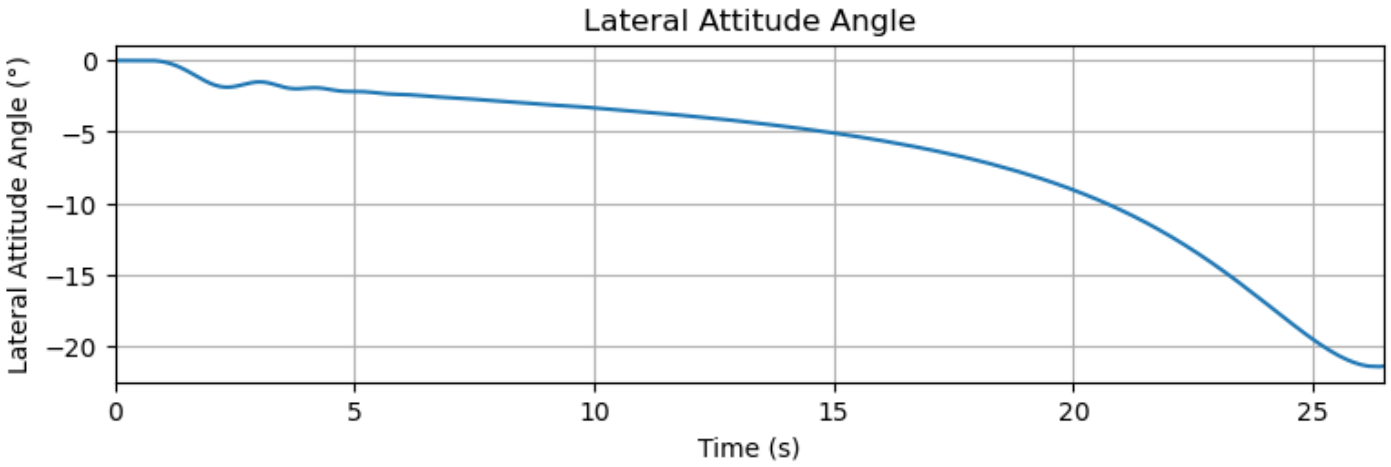



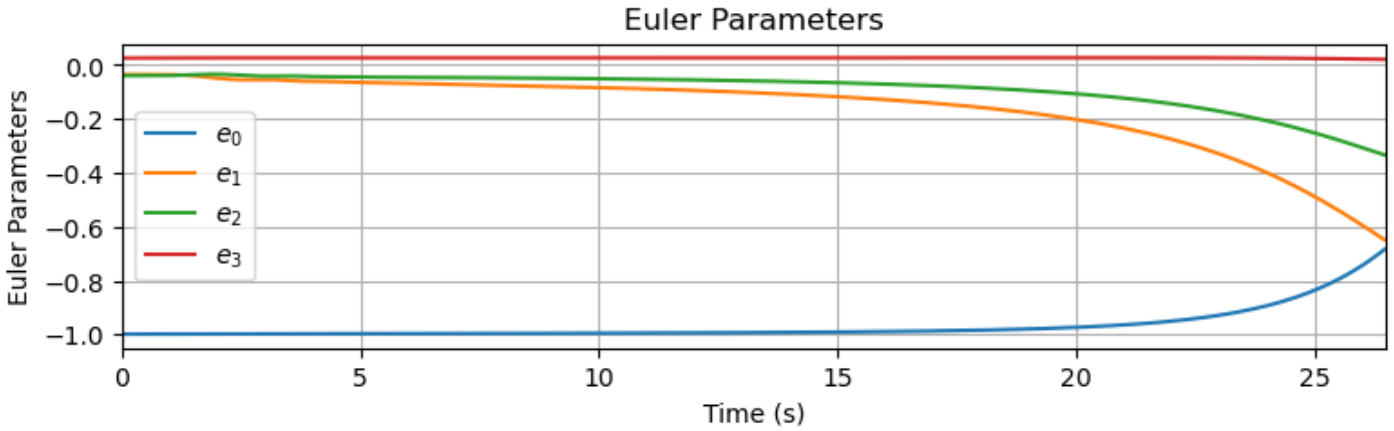



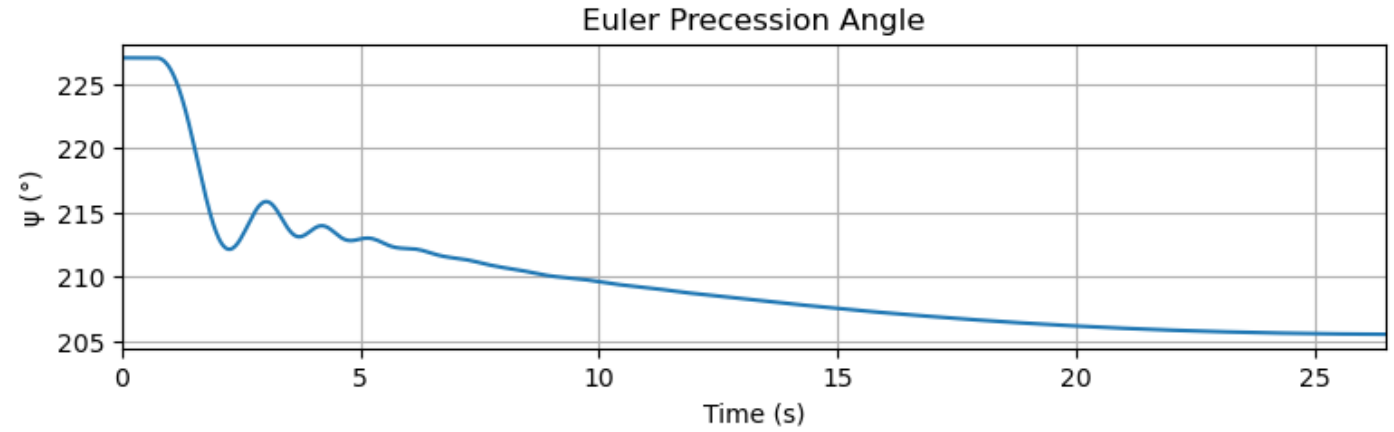

Angular Position Plots

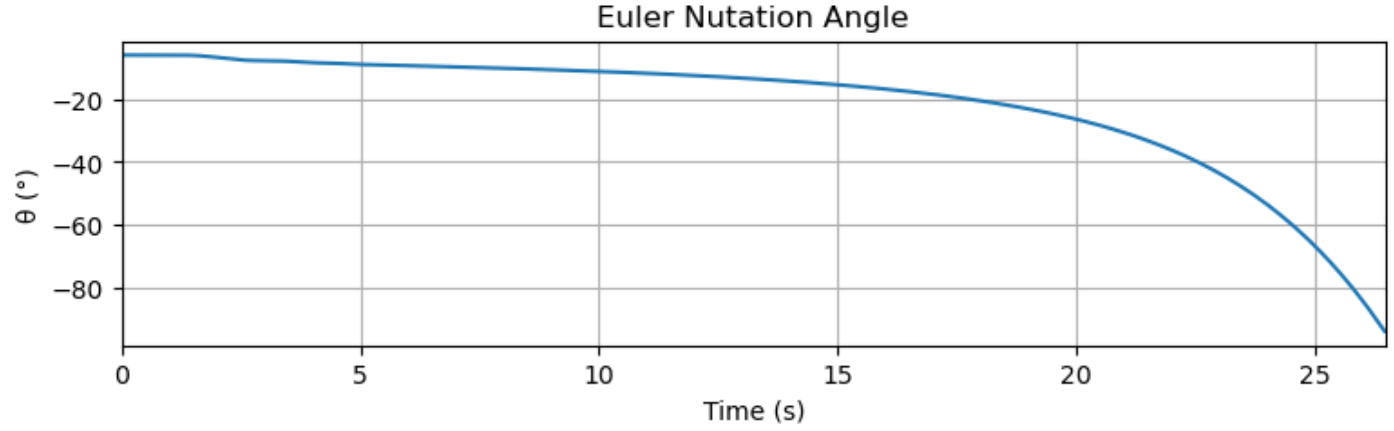


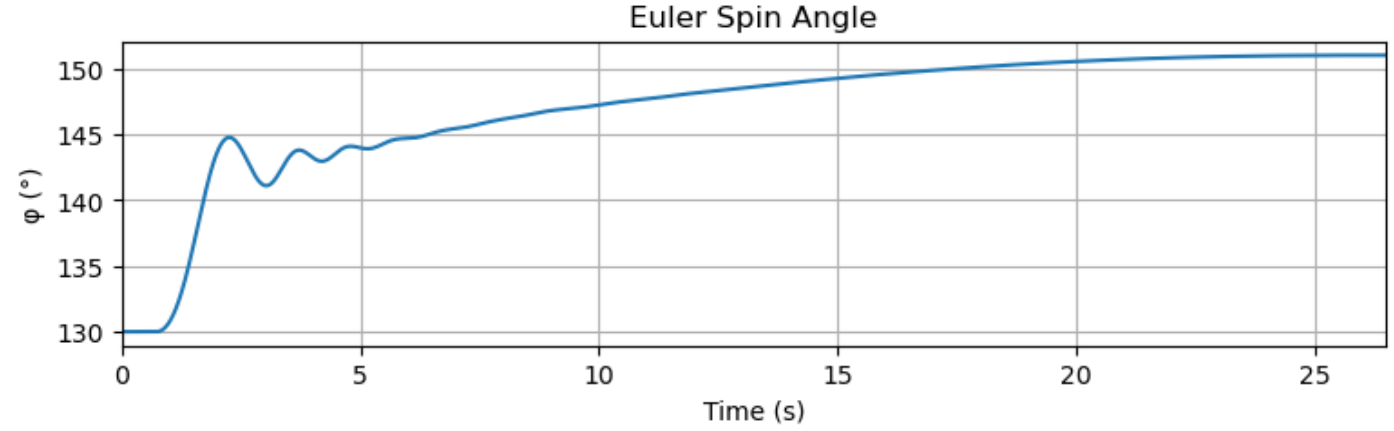




Path, Attitude and Lateral Attitude Angle plots

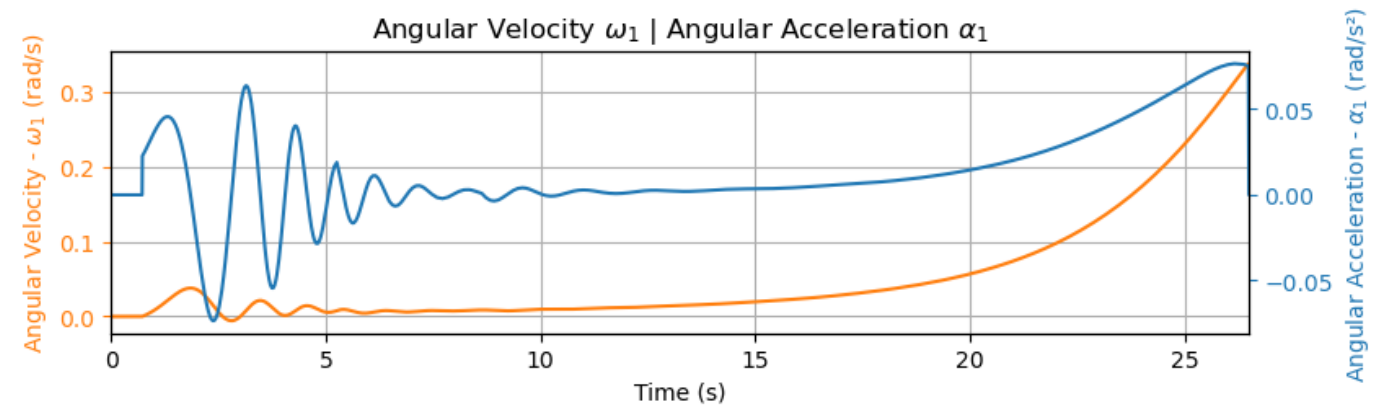


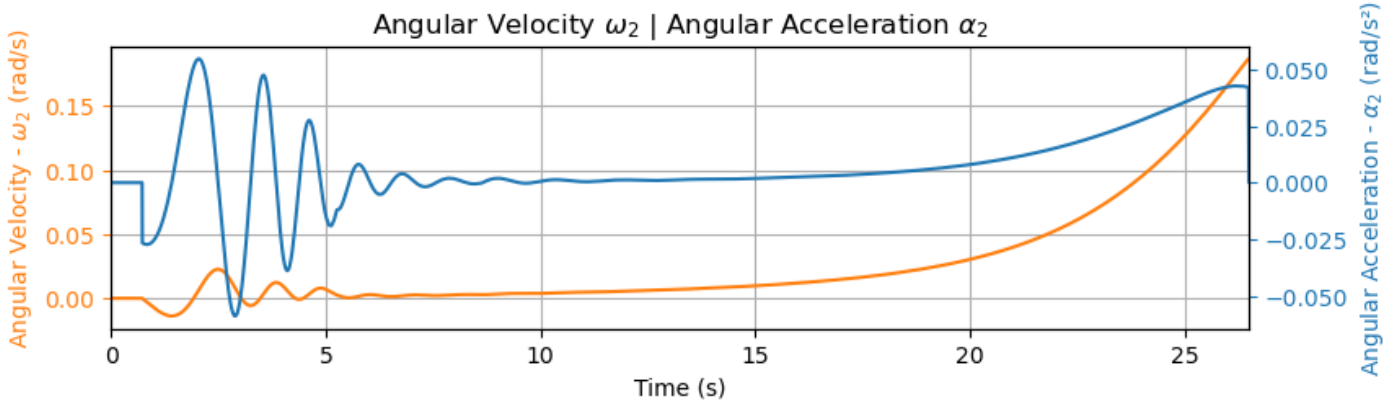



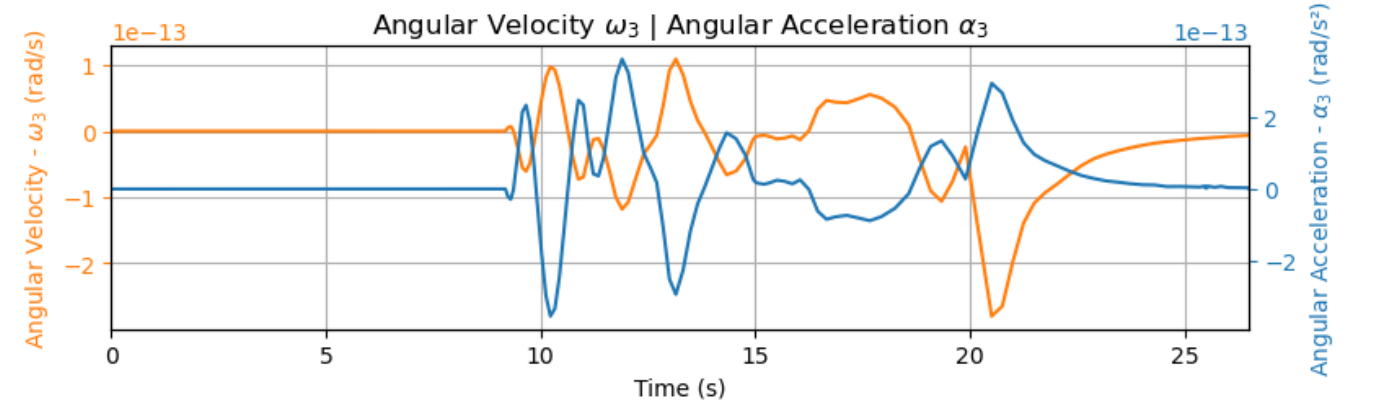



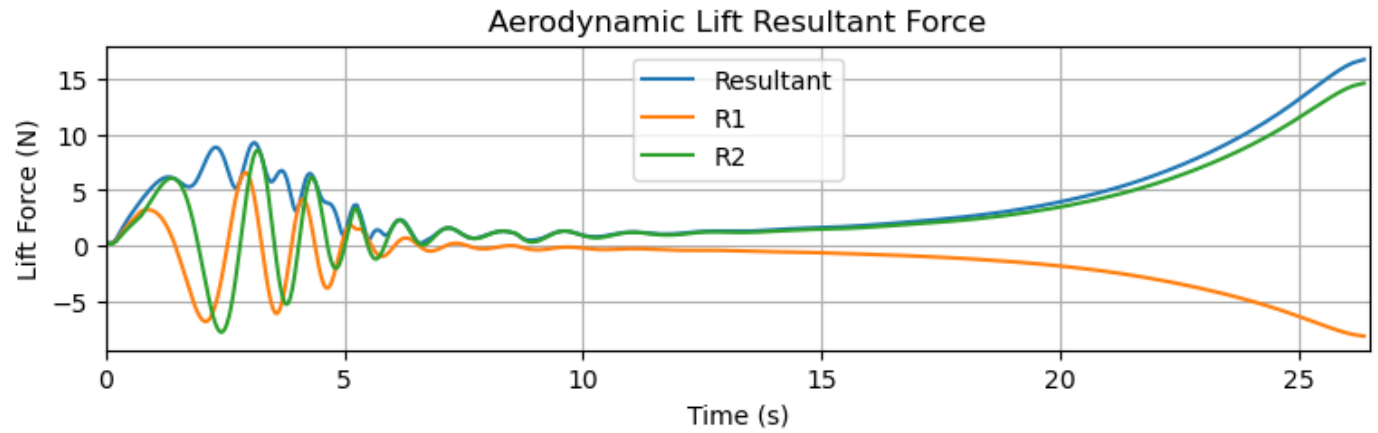

Trajectory Angular Velocity and Acceleration Plots

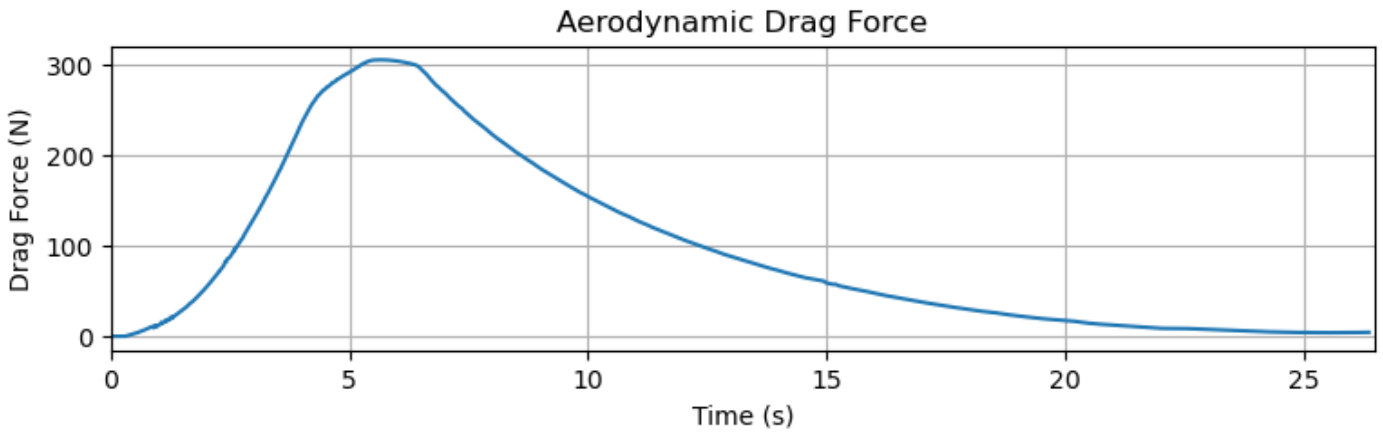


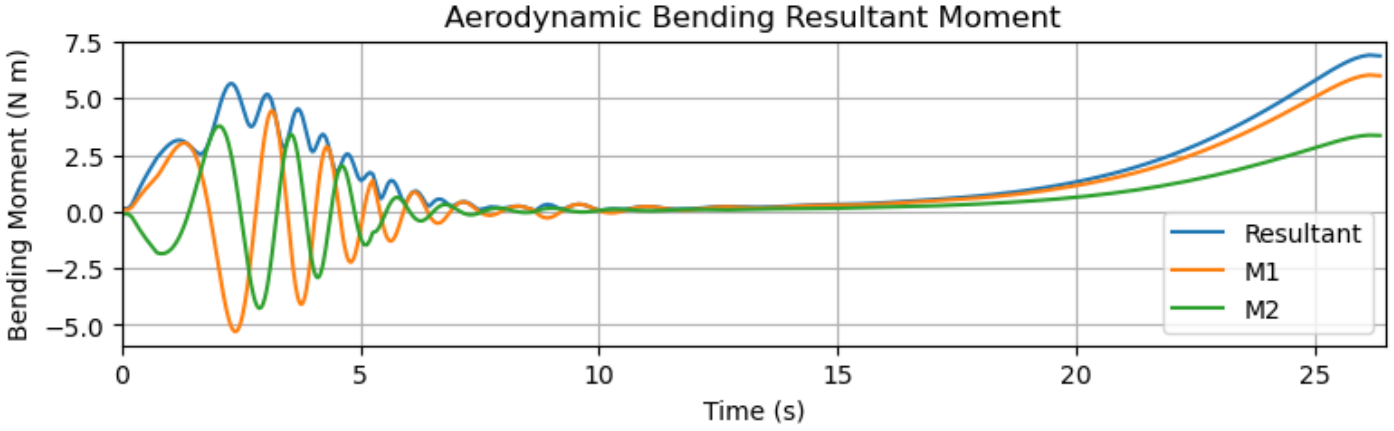



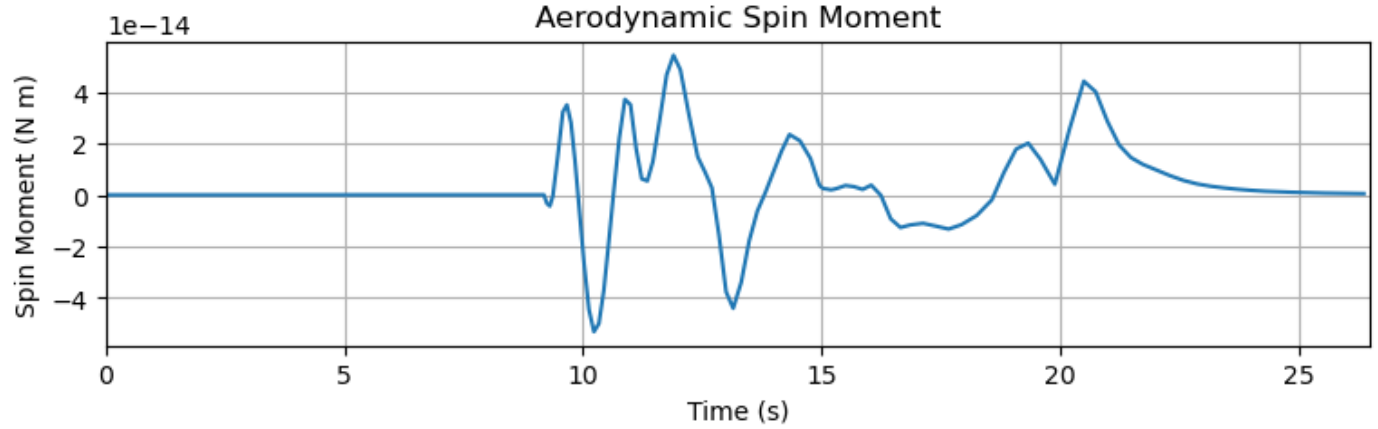




Aerodynamic Forces Plots



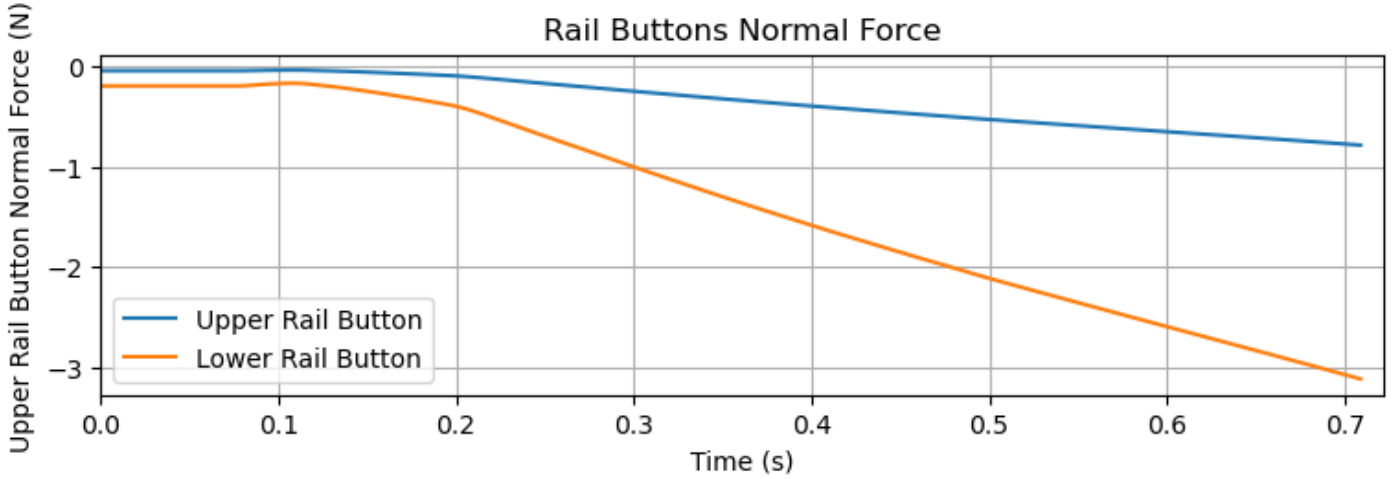



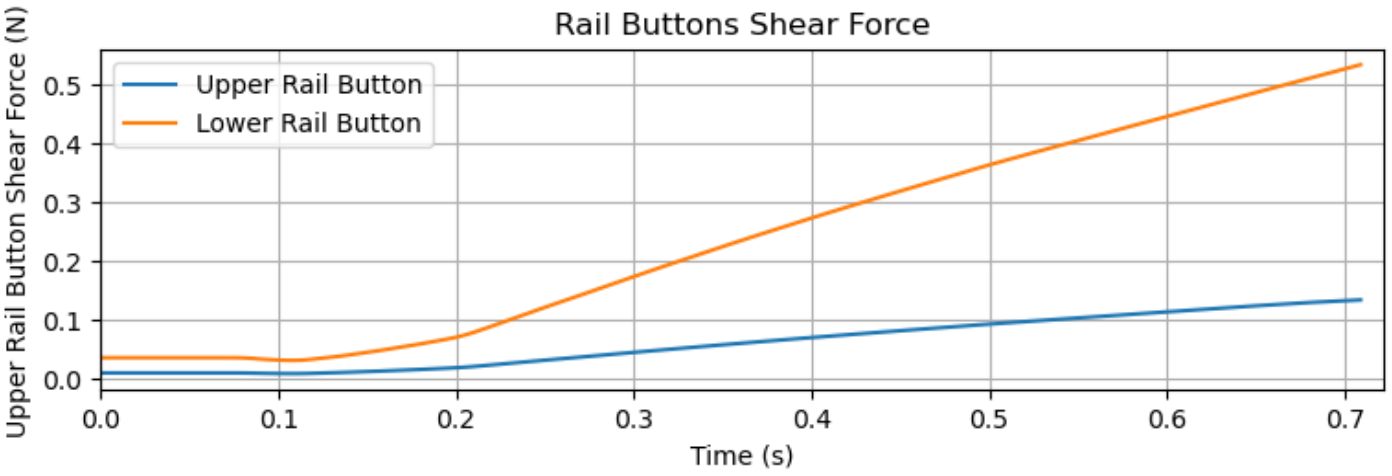




Rail Buttons Bending Moments Plots

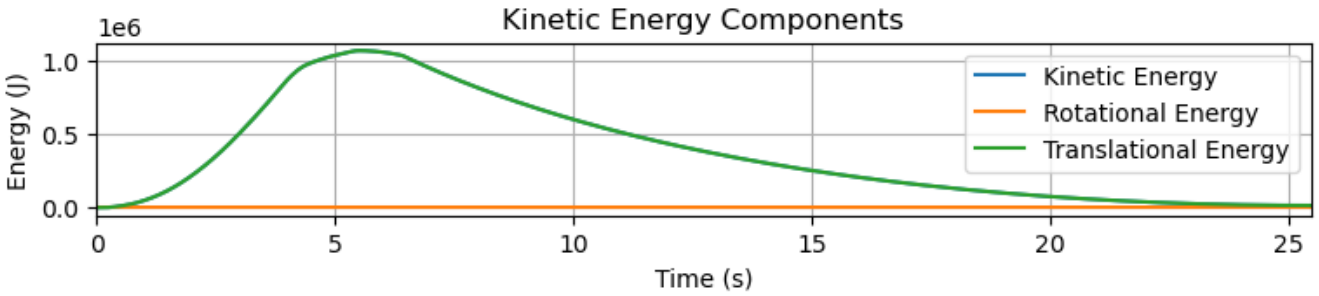
Rail button height not defined. Skipping bending moment plots.

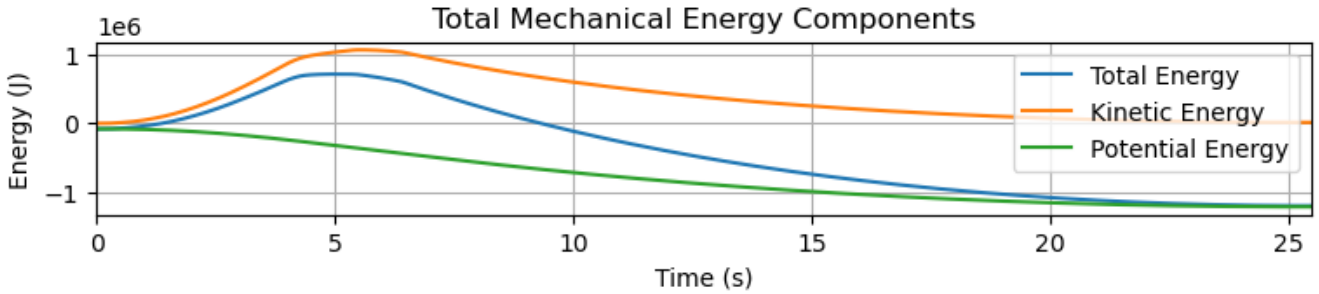
Rail Buttons Forces Plots

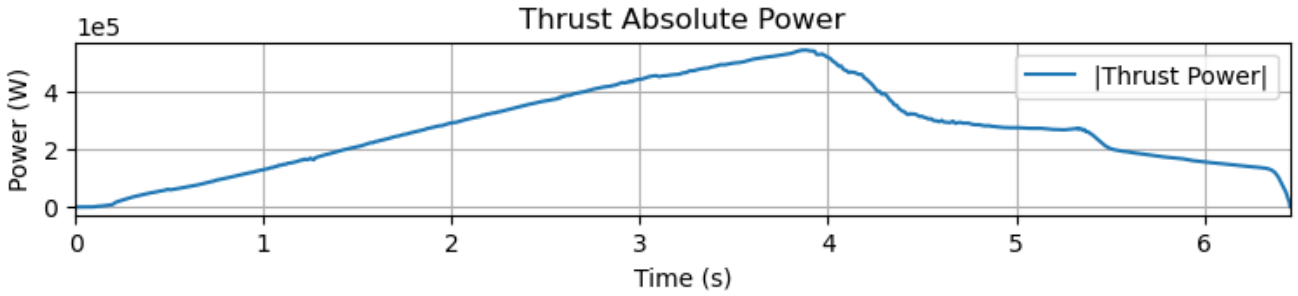


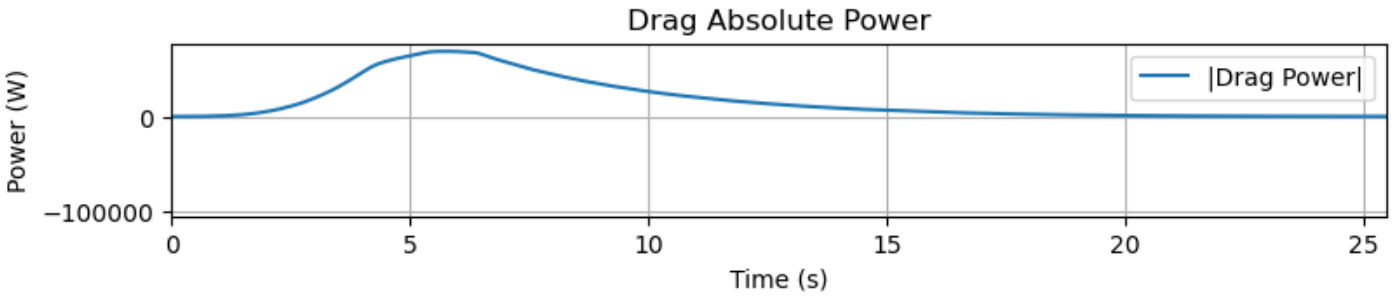




Trajectory Energy Plots

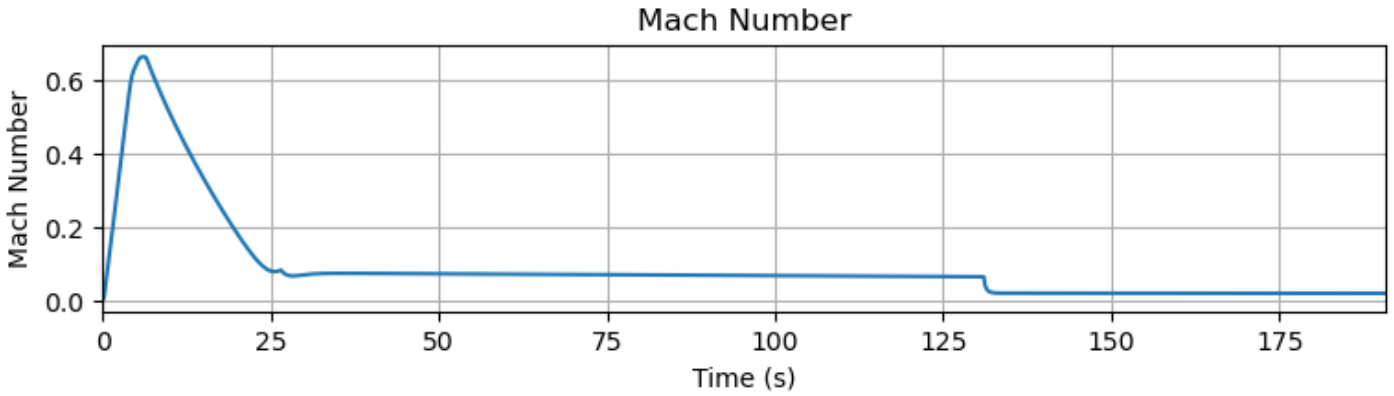


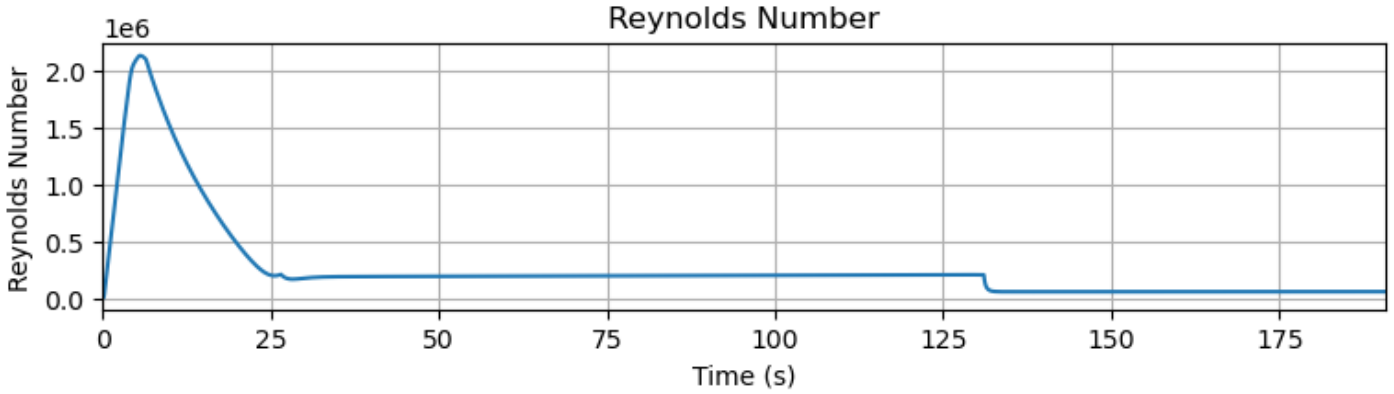



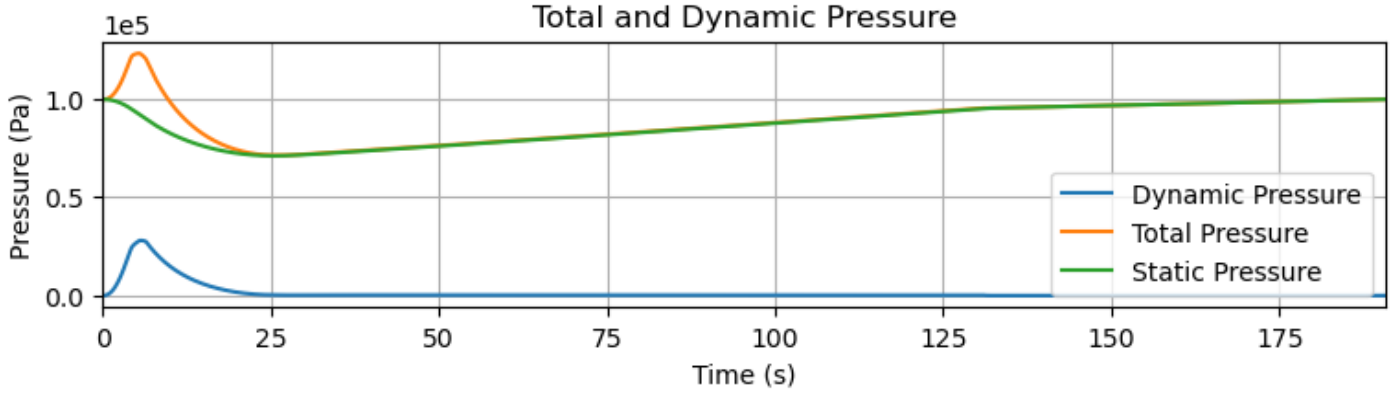



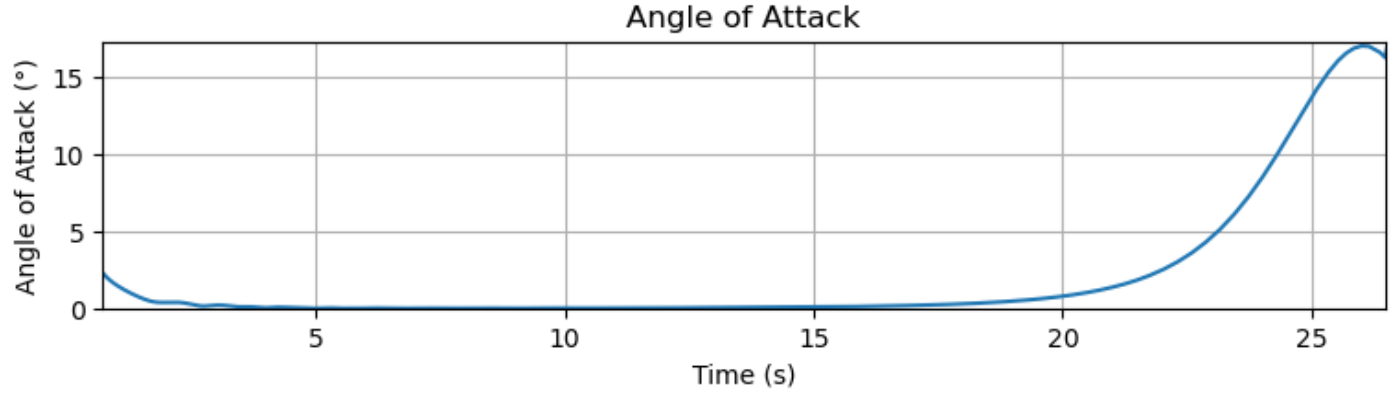



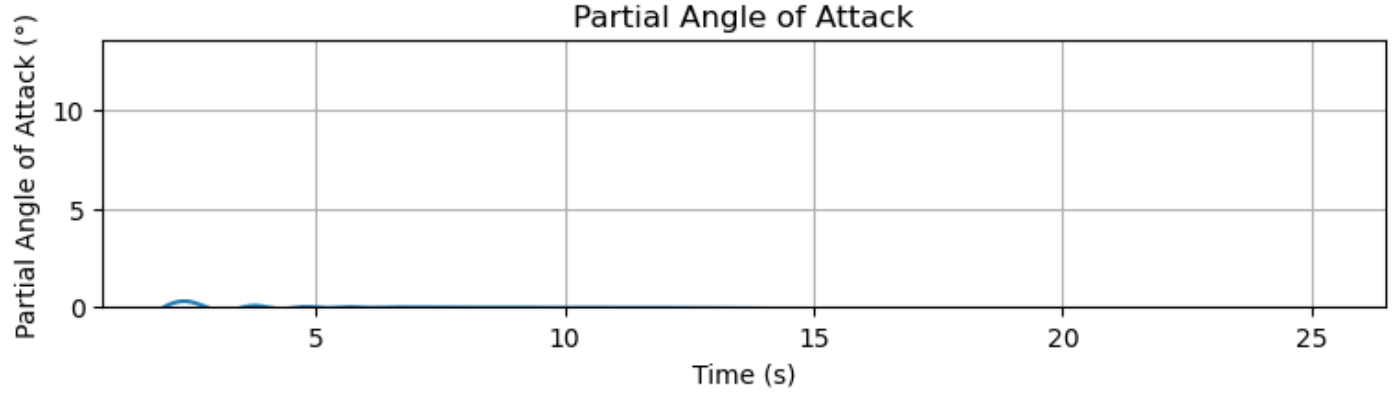

Trajectory Fluid Mechanics Plots

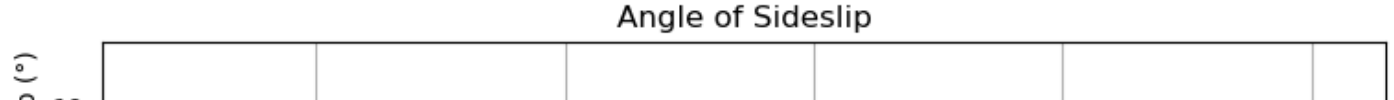

Mach Number
Mach Number
0.0
0.2
0.4
0.6
0
25
50
75
100
125
150
175
Time (s)

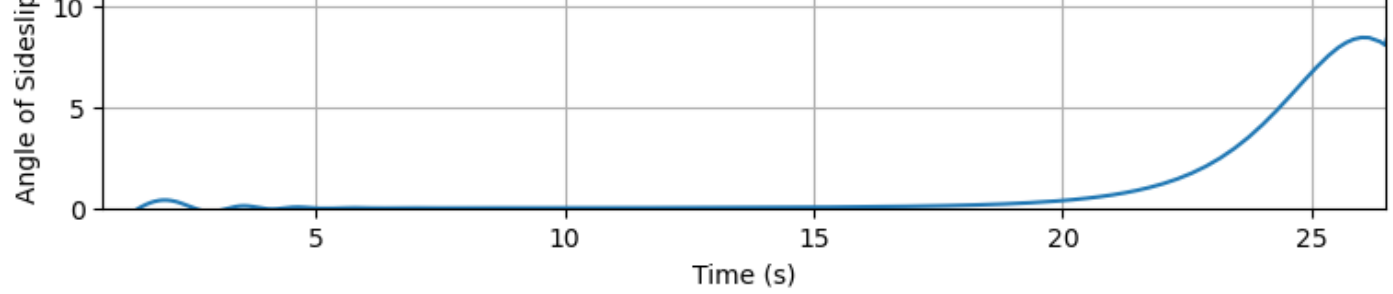

Reynolds Number
1e6
Reynolds Number
0.0
0.5
1.0
1.5
2.0
0
25
50
75
100
125
150
175
Time (s)


Total and Dynamic Pressure
1e5
Pressure (Pa)
0.0
0.5
1.0
0
25
50
75
100
125
150
175
Time (s)
Dynamic Pressure
Total Pressure
Static Pressure

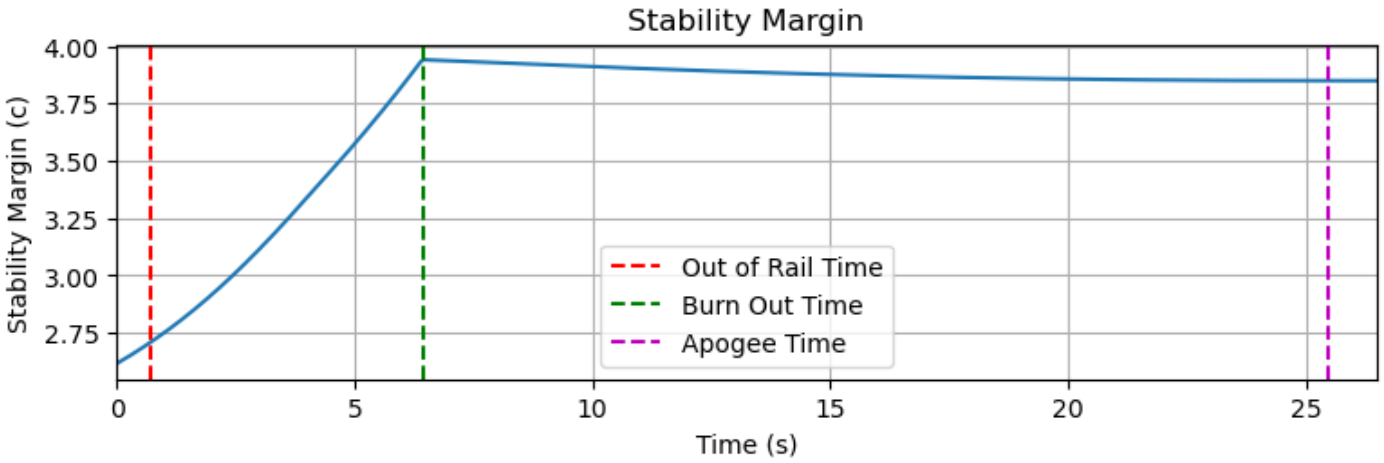

Angle of Attack
Angle of Attack (°)
0
5
10
15
5
10
15
20
25
Time (s)

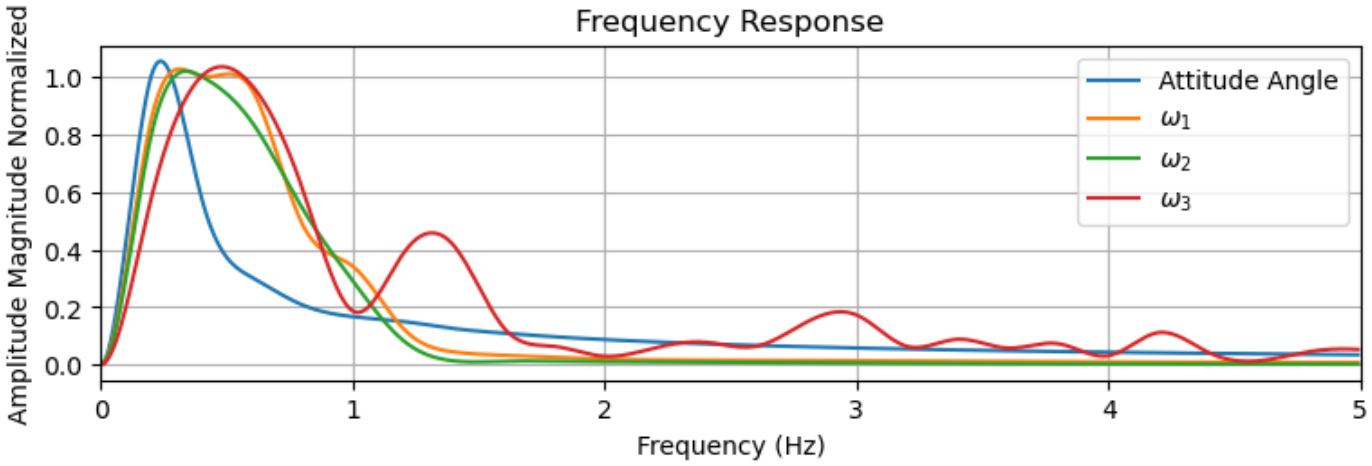

Partial Angle of Attack
Partial Angle of Attack (°)
0
5
10
5
10
15
20
25
Time (s)


Angle of Sideslip

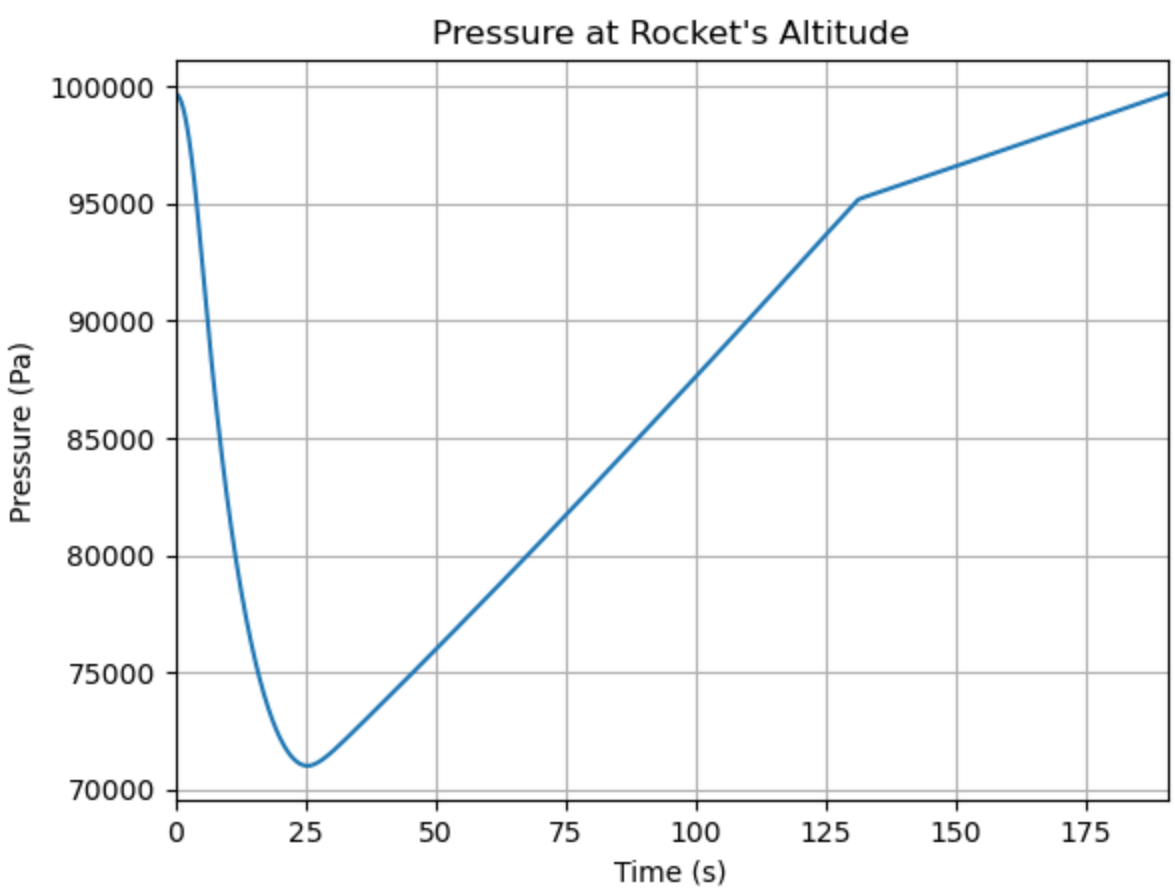



Trajectory Stability and Control Plots

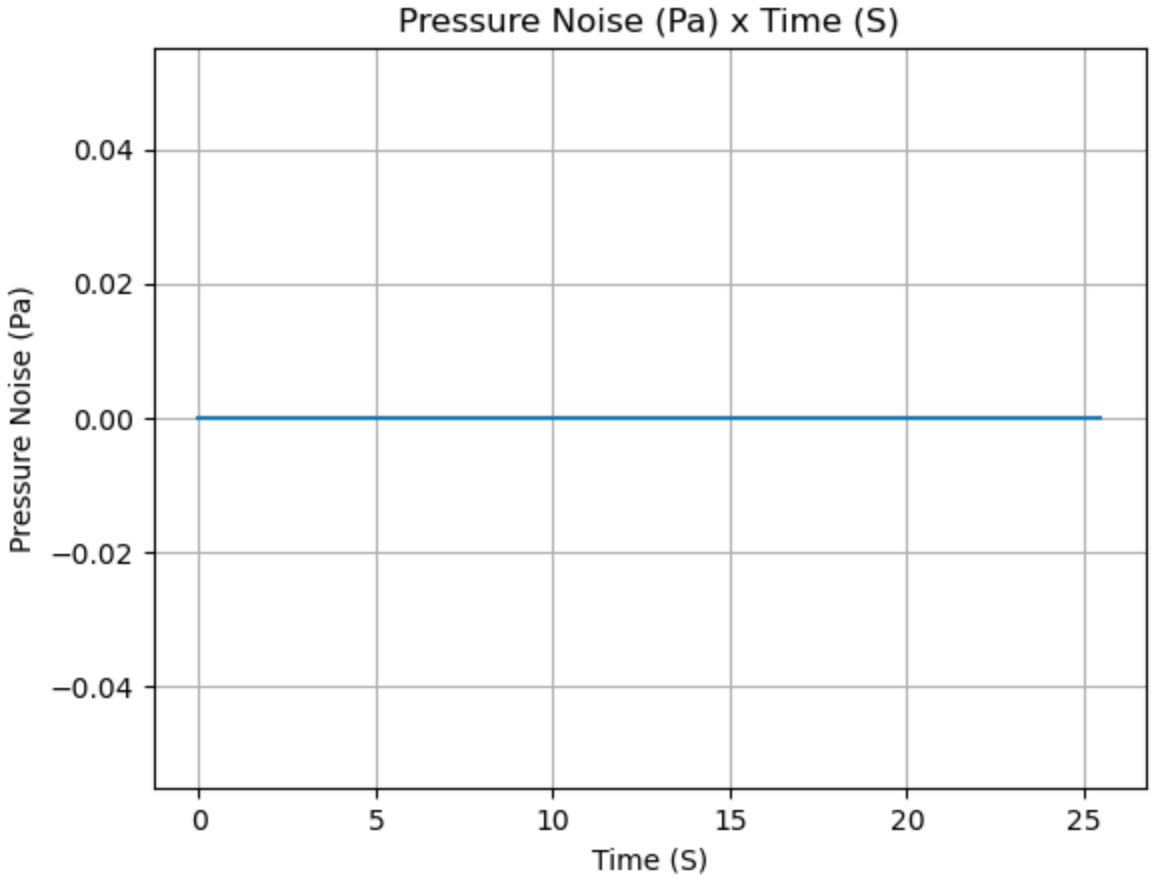


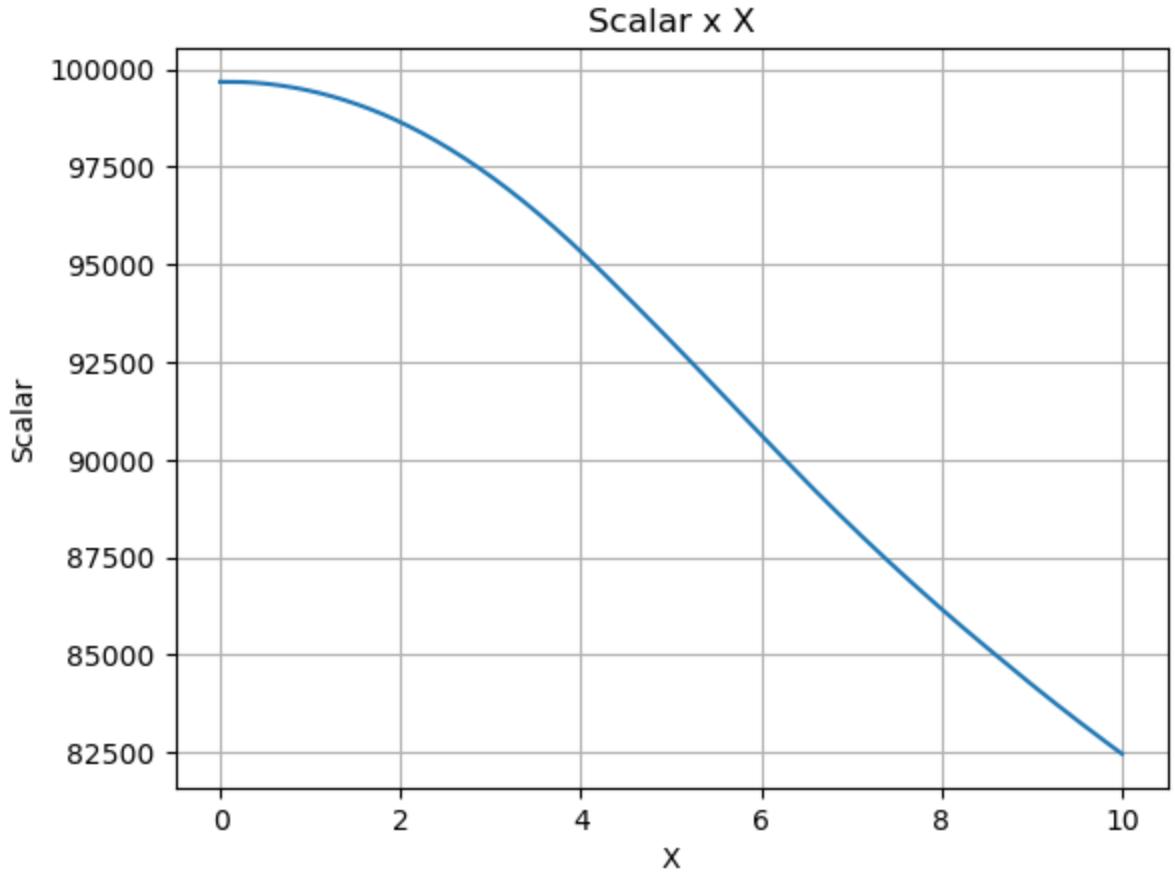



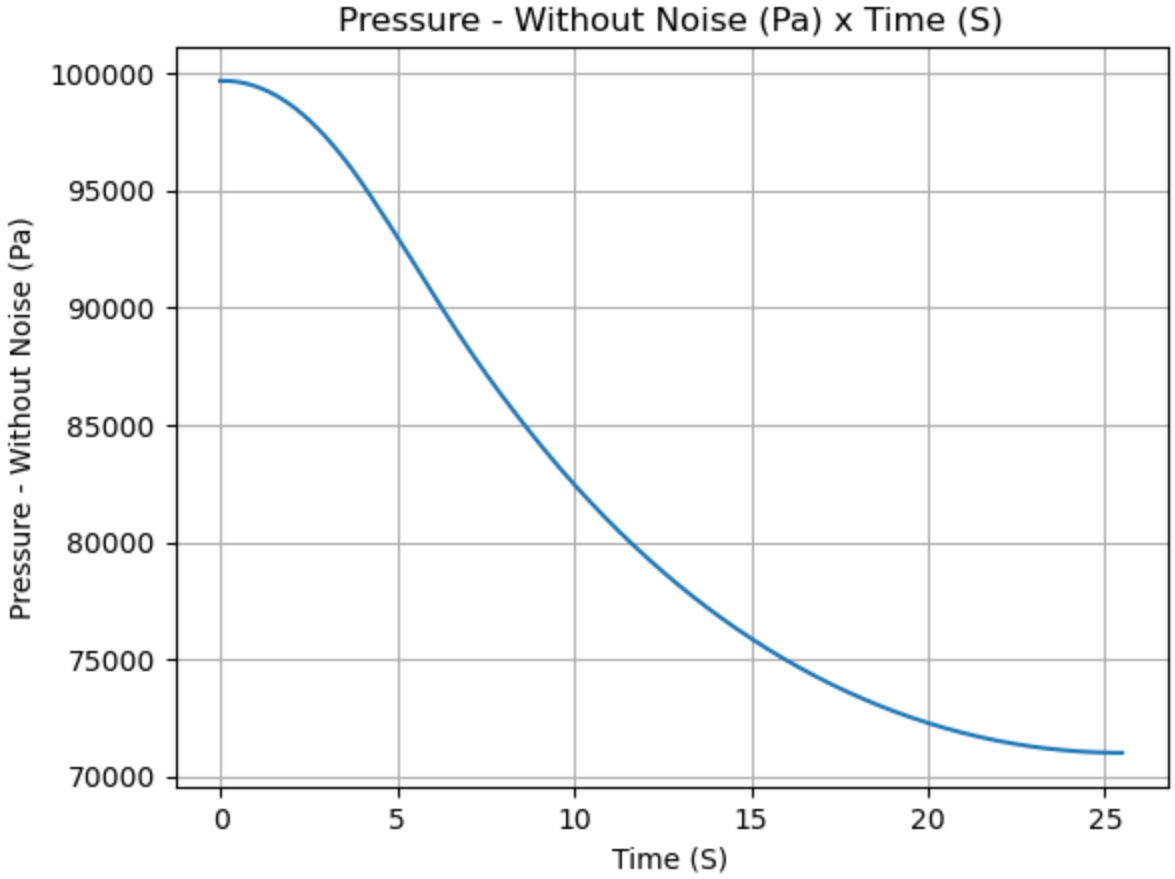

Rocket and Parachute Pressure Plots



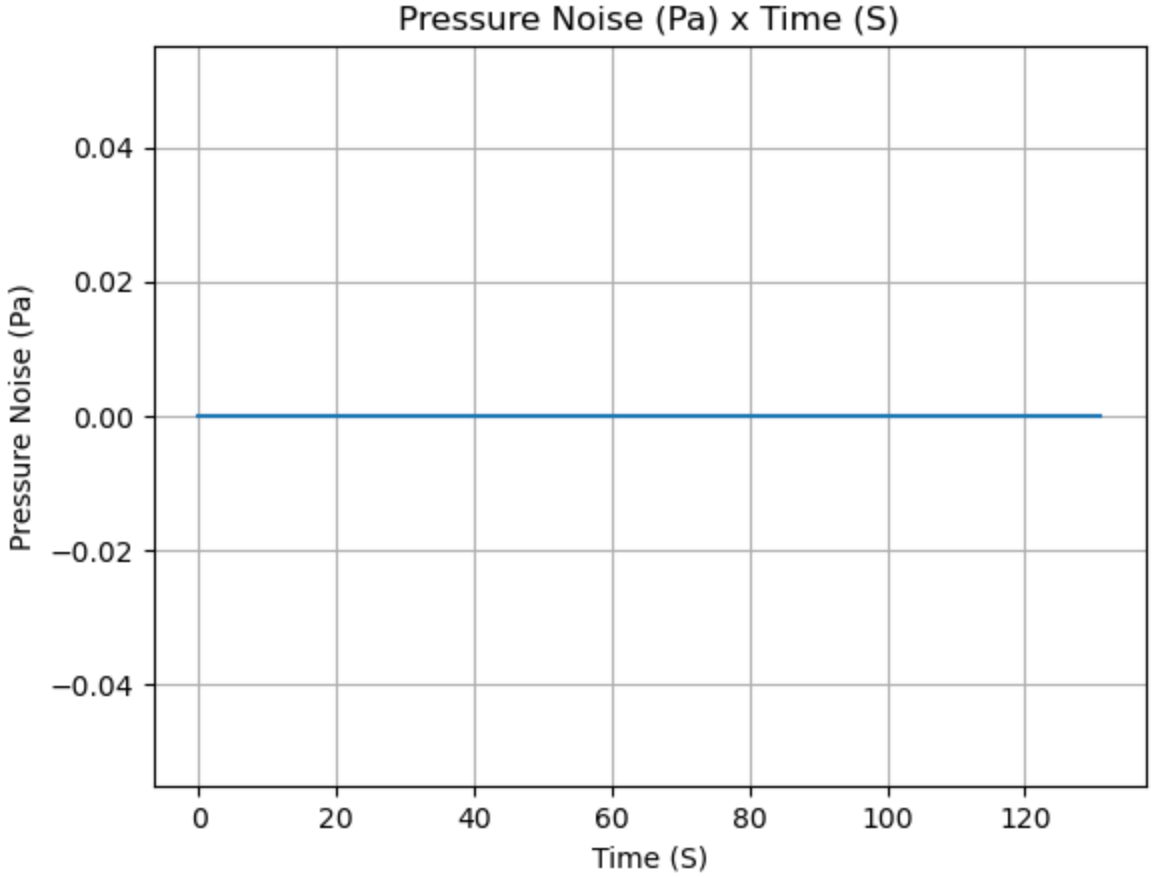

Parachute:  Pilot Parachute

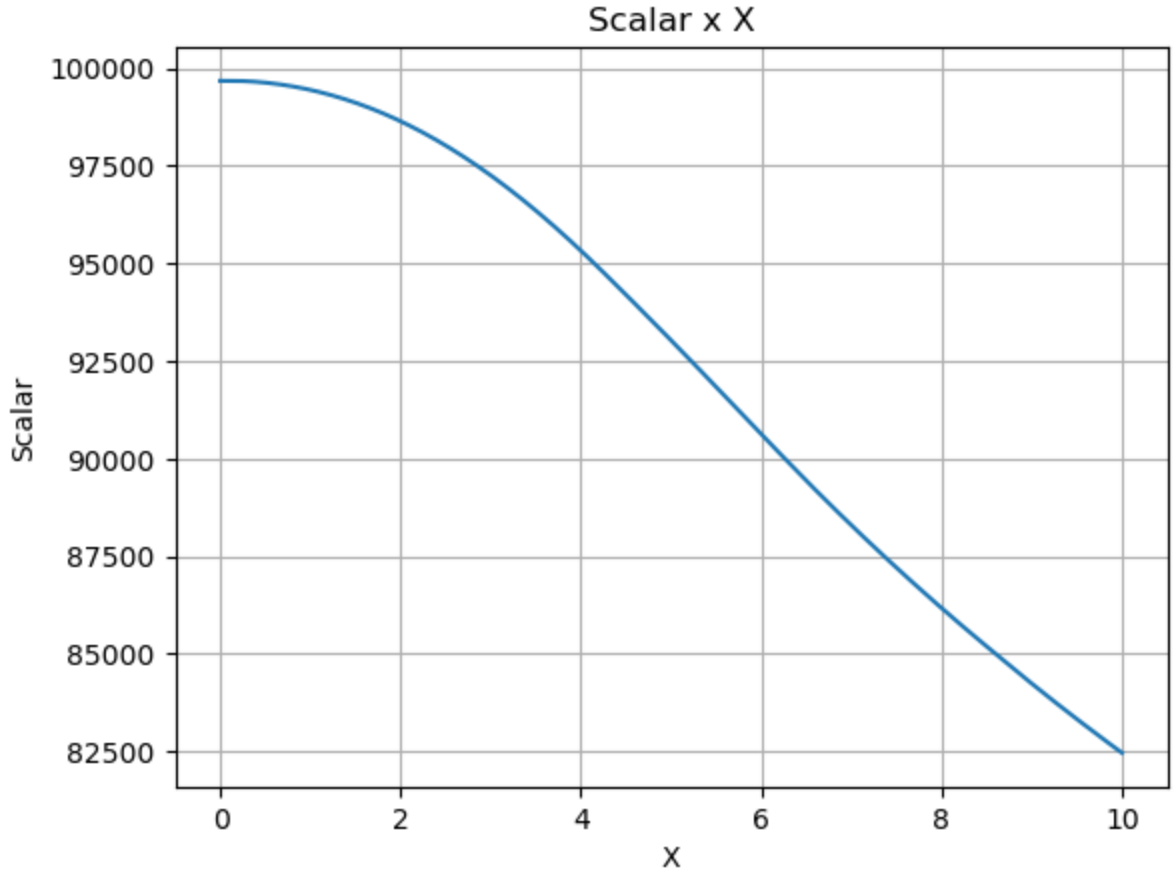


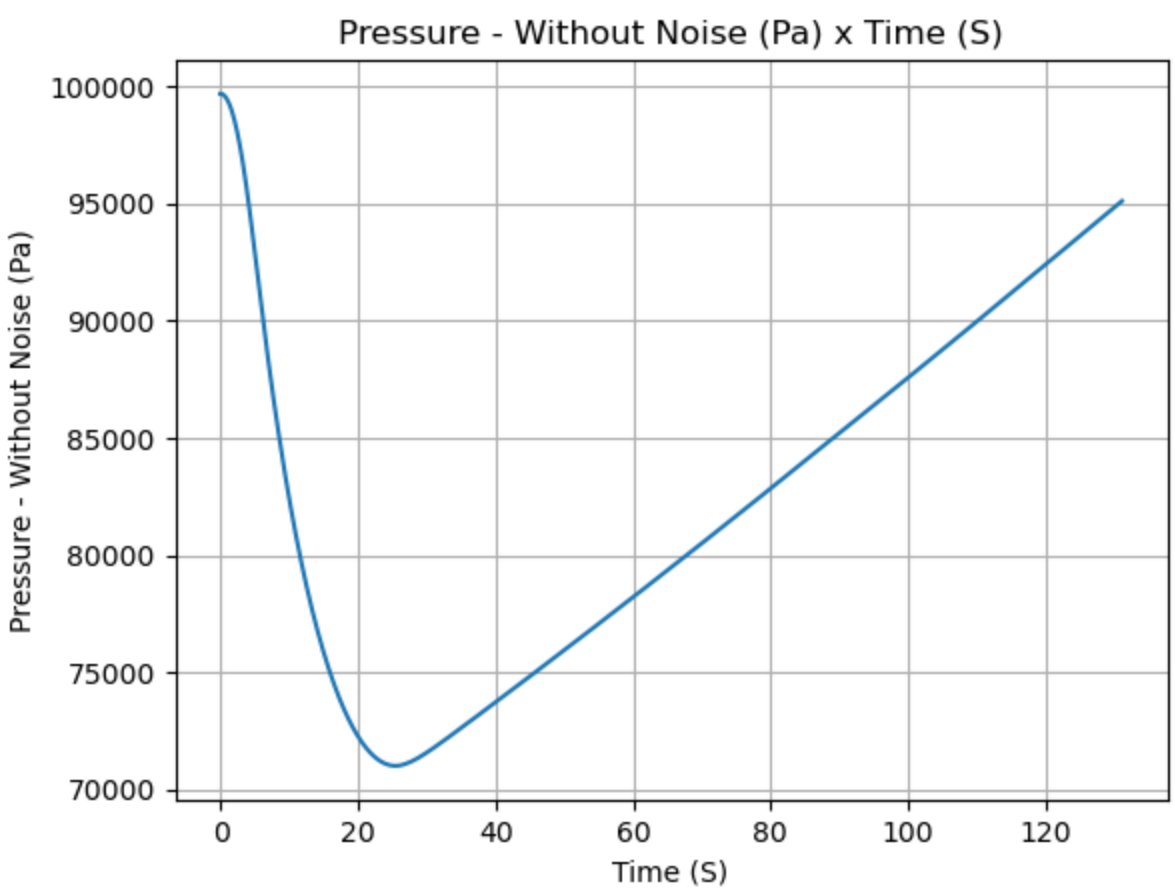



Parachute:  Main Parachute


Pressure Noise (Pa) x Time (S)
Pressure Noise (Pa)
0.04
0.02
0.00
−0.02
−0.04
0
20
40
60
80
100
120
Time (S)


Scalar x X
Scalar
100000
97500
95000
92500
90000
87500
85000
82500
0
2
4
6
8
10
X


Pressure - Without Noise (Pa) x Time (S)
Pressure - Without Noise (Pa)
100000
95000
90000
85000
80000
75000
70000
0
20
40
60
80
100
120
Time (S)